

Yagyaansh Khaneja

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Education

IIT Roorkee, B. Tech. in Eng. Physics (CGPA 7.11/10) <i>First Division</i>	2021
RDPS, Delhi (CBSE), Class XII: PCM (94.2 %)	2017
RDPS, Delhi (CBSE), Class X (CGPA 10/10)	2015

Skills

- Programming:** FORTRAN, Python (PyTorch, NumPy, scikit-learn, pandas, matplotlib), HTML/CSS, SQLite
- Tools:** MATLAB, COMSOL, CST MW Studio, Excel/Word/PowerPoint, LaTeX, Lightroom
- Languages:** English (Native), Hindi (Native), French (Basic), Punjabi (Basic)

Certificates

- Machine Learning:** Andrew Ng ([Coursera](#)) [2023]
- Python for Everybody:** Charles Severance ([Coursera](#) Specialization) [2020]
- Web Design for Everybody:** Colleen ven Lent ([Coursera](#) Specialization) [2020]
- Internet: History, Technology, Security:** Charles Severance ([Coursera](#)) [2020]

Achievements

- NGPE 2019 (Part A):** Top 10%
- JEE 2017 (Advanced):** AIR 3188
- JEE 2017 (Mains):** AIR 2933 (Top **0.2%**)
- Vijyoshi 2016 (Camp):** Invited Guest
- KYPY 2015 (SA):** AIR 300 (Top **0.4%**)
- NTSE 2015 (II):** Scholar (Top **0.1%**)

Physics Projects

- **Bachelor's Project – Silicon Photonic Switch, IIT Roorkee** [May 2020 – Sep 2021]
 - **Organization:** Physics Dept., IIT Roorkee ([Prof. Rajesh Kumar](#))
 - **TL;DR:** Simulated non-volatile GST-SOI waveguides using CST MW Studio and COMSOL.
 - Designed photonic switches for use at 2.1µm ([report](#)) and 1 THz ([thesis](#))
 - Studied Mux-DeMux using Optical Phased Array & WDM for telecom applications
 - **Skills:** EM Simulation ([COMSOL](#), CST MW Studio), MATLAB, Waveguide Photonics
- **Research Camp – Optics/Photonics, ITMO University** [May 2019 – Jun 2019]
 - **Organization:** ITMO University, St. Petersburg, Russia ([Prof. Anton Tcypkin](#))
 - **TL;DR:** Gained exposure to advances in Optics through [active engagement](#) with frontier research ([Certificate](#))
 - Generated THz pulse radiation via laser filamentation of liquids ([First global attempt](#) using pH variation of water!)
 - Simulated interference & diffraction patterns for varying waveforms & apertures
 - Studied Quantum Information, Holography, QKD Systems, Femtosecond Lasers, etc.
 - **Skills:** Experimental Optics, MATLAB, Laser Systems

Tech Projects

- **Project Manager (Volunteer) – Tech Team** [Dec 2024 – Jun 2025]
 - **Organizations involved:** [Tarayana](#), [GIAOS](#), [Omniverse](#)
 - **TL;DR:** Guided the development a [data dashboard](#) (proof-of-concept) for a remote village in Bhutan
 - Under the aegis of renowned banker Tan Sri [Andrew Sheng](#) SBS
 - **Skills:** Team Management, High Level Architecture Design, Tech Stack: React Native, MongoDB, DeepSeek
- **Research Intern (Energy Team) – Sonam Wangchuk** [Jan 2022 – May 2022]
 - **Organization:** Himalayan Institute of Alternatives ([HIAL](#)), Ladakh
 - **TL;DR:** Assisted renowned environmentalist [Sonam Wangchuk](#) on his projects (SDGs)
 - Energy Analysis of PCM in Trombé Walls (Passive Solar Heated), and HVAC systems
 - Drafted [manual](#) for construction of Passive Solar Heated ([PSH](#)) Buildings in Ladakh.
 - Maintained and installed solar PV systems (20 kW, 7.5 kW)
 - Installed new Stevenson Screen; assisted in weather [data collection](#) (& analysis) in remote regions of Ladakh using SensorPush and Campbell Scientific weather stations.
 - **Skills:** Research (Thermal Physics), Excel, Project Implementation, Electrical Science

Misc & Volunteering

- STEM & English Teacher** (May-Aug, 2022)
 - **Locations:** Ladakh ([SECMOL](#)), Vietnam ([Workaway](#))
 - Taught dropouts using hands-on science/maths [experiments](#).
 - Led **astronomy** and photography hobby clubs.
 - **Skills:** Communication, Empathy
- Leadership Roles, IIT Roorkee** (2017-2021)
 - Additional Secretary, Photography Section
 - Coordinator, Thomso (Cultural Fest)
 - Content Manager, Srishti (Tech Exhibition)
 - **Skills:** Leadership, Management

Work Experience:

- **Family Business – B2B Wholesale (pan India)** [Sep 2022 – Sep 2025]
 - **Location:** Asia's largest wholesale market: Chandni Chowk/Chawri Bazaar (Delhi-8)
 - **TL;DR:** In essence, increased the efficiency of operations, **saving 4 hours per day**
 - Migrated **accounts** from legacy (handwritten) to cloud software, enabling remote work for the first time in 35 years.
 - Meticulously classified 1000s of SKUs (**Steel products**), 400+ parties; making data entry organized, eliminating the need for “data cleaning” while analysis.
 - Cut operational overhead via warehouse optimization (Streamlined packaging, labels, stickers, colour coded dispatch, item codes, and process flows).
 - Initiated **online retail** sales through [Mecsho](#), expanding customer base.
 - Designed a [catalog](#) of Door Fittings, Fasteners, Screws - simplifying products for customers
 - Learned invaluable lessons about Money, **Cash Flow**, Relationship Management, Business Ethics, Sales, etc.
 - **Skills:** Warehouse Management, Leadership, Canva Design, Accounting & Financial Management

Other Cool Stuff

- **Standard Model - Visualized** [May 2025 – ongoing]
- TL;DR: Visualized Standard Model of Particle Physics in 3D
 - Used three properties at a time (mass, charge, isospin/strangeness)
 - Tried matplotlib initially, then switched to Manim for better animations
- Skills: Python - matplotlib, Manim (by 3Blue1Brown)

- **Personal Website** – yagyaansh.com [Nov 2020 – Jan 2021]
- TL;DR: Hard-coded my website using basic HTML, CSS and JavaScript (no frameworks)
 - Implemented Dark Theme toggle using JavaScript
 - Wrote blogs with graphics, integrated comments section using Disqus
 - Hosted my website using GitHub Pages
- Skills: Web Dev (HTML, CSS, JS), handling SVG in HTML

- **Fee Data Extraction** [Aug 2020]
- TL;DR: Made API calls (Azure server) to extract data of 3000+ students (college fee receipts)
 - Libraries used - urllib.request, ssl, json, sqlite3 in Python
 - converted the received json data into a SQLite database for easy accessibility
- Skills: Handling JSON and SQLite database using Python

Coursework (B. Tech.)

- **Basics**
 - **EMT**: Sadiku; Griffiths
 - **Relativity**: Resnick; Goldstein; Aurther Beiser
 - **QM 1**: Griffiths; Zettili
- **Maths**
 - **Mathematical Physics**: Mary L. Boas; Arfken
 - **Advanced Maths**: Kreyszig; Brown & Churchill
 - **Computational Physics**: DeVries (FORTRAN)
- **Nuclear**
 - **Nuclear Physics**: Krane; John Lilley
 - **Nuclear Astrophysics**: Rolfs & Rodney; Iliadis
- **Interdisciplinary**
 - **Thermal & Statistical Physics**: Sears & Salinger; Zemansky & Dittman; Landau & Lifshitz
 - **Atomic & Molecular Physics**: Haken & Wolf; Bransden
 - **Condensed Matter Physics**: Wahab
 - **Plasma Physics**: Francis F. Chen
- **Electronics**
 - **Electronics**: Malvino; Millman; Streetman; Horowitz; Scherz
 - **Electrical**: Chapman; Hughes
 - **Electrical & Electronic Materials**: S. O. Kasap
 - **Microprocessors**: Ramesh S. Gaonkar
 - **Semiconductor Devices**: Sze & Lee; Streetman
- **Optics/Photonics**
 - **Applied Optics**: Hecht; Born & Wolf; Ghatak
 - **Lasers & Photonics**: Ghatak & Thyagarajan; Silfvast; Svelto; Yariv
 - **Optical Communication**: G. P. Agrawal; Wartak; B. P. Lathi; Schaum
 - **Fiber & Non-Linear Optics**: Ghatak & Thyagarajan
 - **Optoelectronics**: Saleh & Teich; Kasap
 - **Photonic Sensors** (Plasmonics) : Sachin Srivastava; Maeir